

Bluestock Mutual Fund Analytics

Capstone Project Final Report

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Executive Summary

This project delivers a comprehensive end-to-end data pipeline and analytics solution for the Bluestock Mutual Fund portfolio. We successfully ingested, cleaned, and integrated 10 disparate raw CSV datasets into a highly optimized SQLite Star Schema database. Leveraging this foundation, we conducted extensive Exploratory Data Analysis (EDA) across 40 mutual fund schemes, covering AUM growth, NAV trends, and investor demographics. Furthermore, we calculated advanced performance and risk metrics (Sharpe, Sortino, Alpha, Beta, VaR, CVaR) and benchmarked funds against NIFTY 50 and NIFTY 100 indices. Finally, we developed a composite Fund Scorecard and a cohort-based behavioral model to guide business decisions.

Problem Statement & Objectives

Objective: To transform fragmented mutual fund data into actionable financial intelligence.

Key Deliverables Achieved:

- Automated ETL Pipeline for accurate data ingestion and cleansing.
- Relational Database Design (Star Schema) enabling robust querying.
- Extensive EDA identifying key market and demographic trends.
- Advanced Risk & Performance Analytics including VaR, Drawdowns, and Benchmarking.
- A unified Fund Scorecard to objectively rank 40 funds.

Data Sources & ETL Pipeline Design

Data Sources:

The project integrated 10 core datasets containing historical NAVs, Benchmark Indices, Fund Master details, SIP flows, AUM records, and investor transactions. The coverage spanned over multiple years with high-frequency daily NAV trading data.

ETL Pipeline Design:

1. Extraction: Raw CSV files were read using Pandas.
2. Transformation: Handled missing values, standardized date formats, calculated forward-filled NAVs (while explicitly flagging non-trading days to preserve return distribution accuracy), and normalized category texts.
3. Loading: Data was pushed into a structured SQLite database (`bluestock\_mf.db`).

Database Design (Star Schema)

The database was architected using a Star Schema optimized for analytics:

- Fact Tables: `fact\_nav` (64,320 rows), `fact\_performance`, `fact\_transactions` (32,778 rows), `fact\_benchmark\_indices`, `fact\_sip\_inflows`, `fact\_aum`, `fact\_holdings`.
- Dimension Tables: `dim\_fund` (40 schemes), `dim\_investor`.

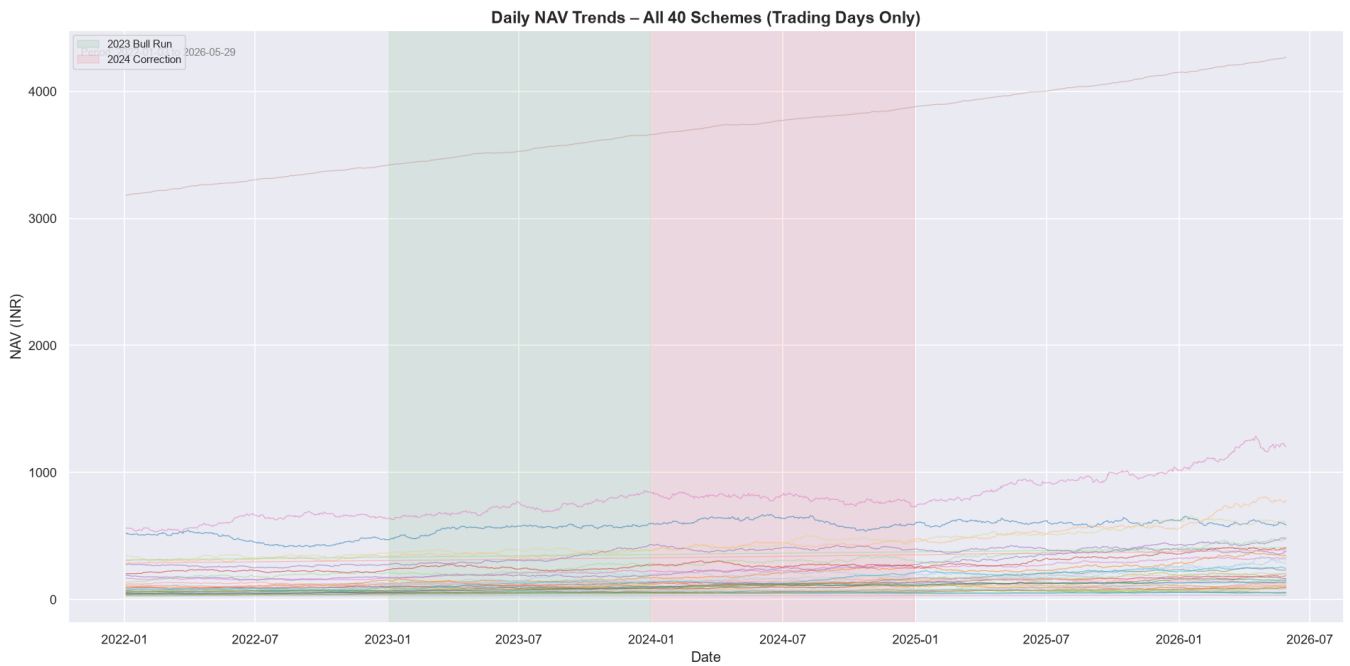
This centralized architecture allowed seamless joins for computing complex metrics like the rolling Sharpe ratio and Alpha.

Exploratory Data Analysis (EDA)

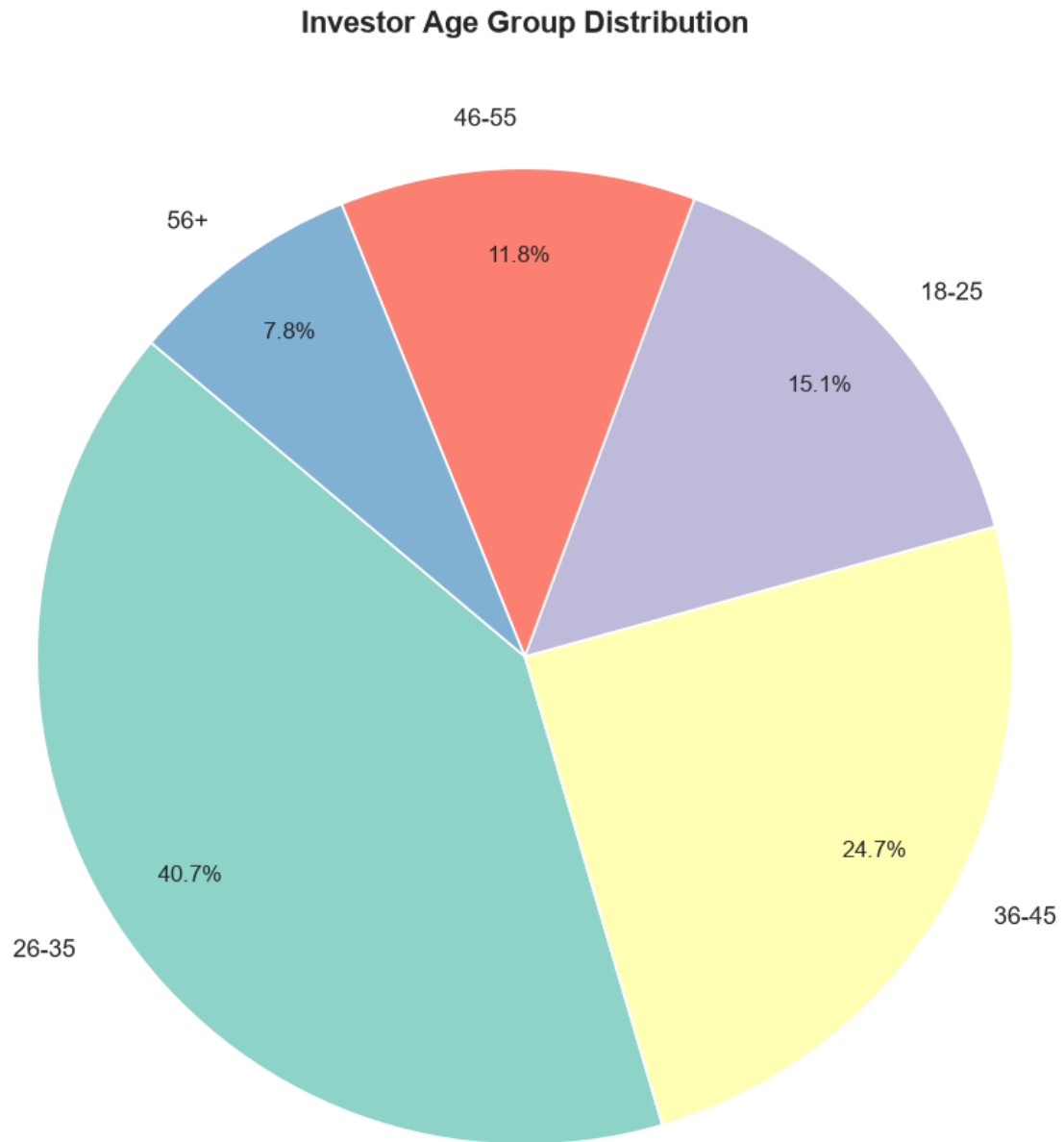
We generated 25 distinct visualizations analyzing fund metrics and demographic trends. Key findings include:

- SBI Mutual Fund consistently holds the highest AUM.
- The '26-35' age group represents 40.7% of all unique investors, dominating the demographic spread.
- T30 cities account for 66.7% of investors, confirming a strong urban penetration.
- High correlation (>0.8) exists amongst the Top 10 equity funds, suggesting limited diversification benefits.

EDA Visualization: 01_nav_trends_all_funds



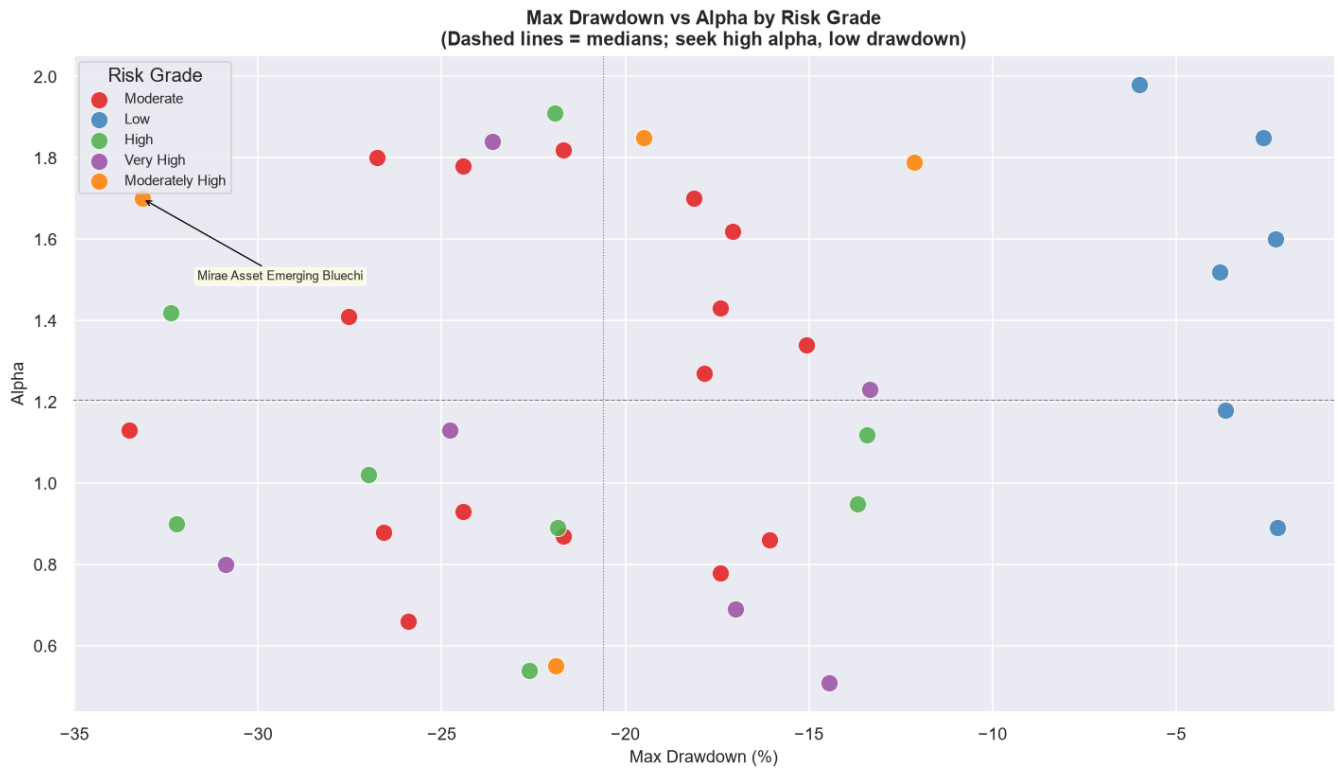
EDA Visualization: 06_age_distribution



EDA Visualization: 18_risk_return_scatter



EDA Visualization: 20_drawdown_vs_alpha

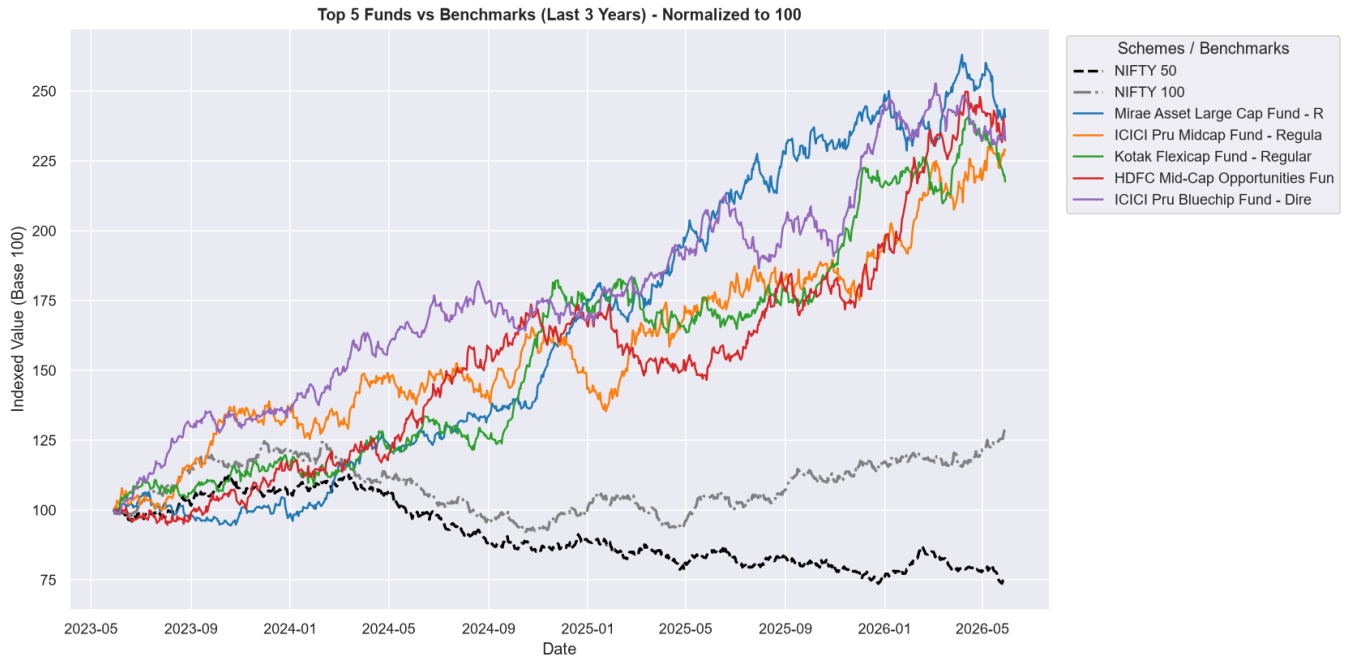


Performance Analytics

To evaluate true fund quality, we computed the following across all 40 funds:

- Daily Returns & CAGR (1yr, 3yr, 5yr).
- Sharpe Ratio & Sortino Ratio: Calculated using an annualized Risk-Free rate of 6.5%.
- Alpha and Beta: Computed via OLS Regression against the NIFTY 100 benchmark.
- Maximum Drawdown: Extracted the worst peak-to-trough decline dynamically.
- Fund Scorecard: A composite 0-100 score prioritizing Risk-Adjusted Returns (Sharpe 25%, Alpha 20%) over raw returns.

Top 5 Funds vs Benchmarks (NIFTY 50 & 100)

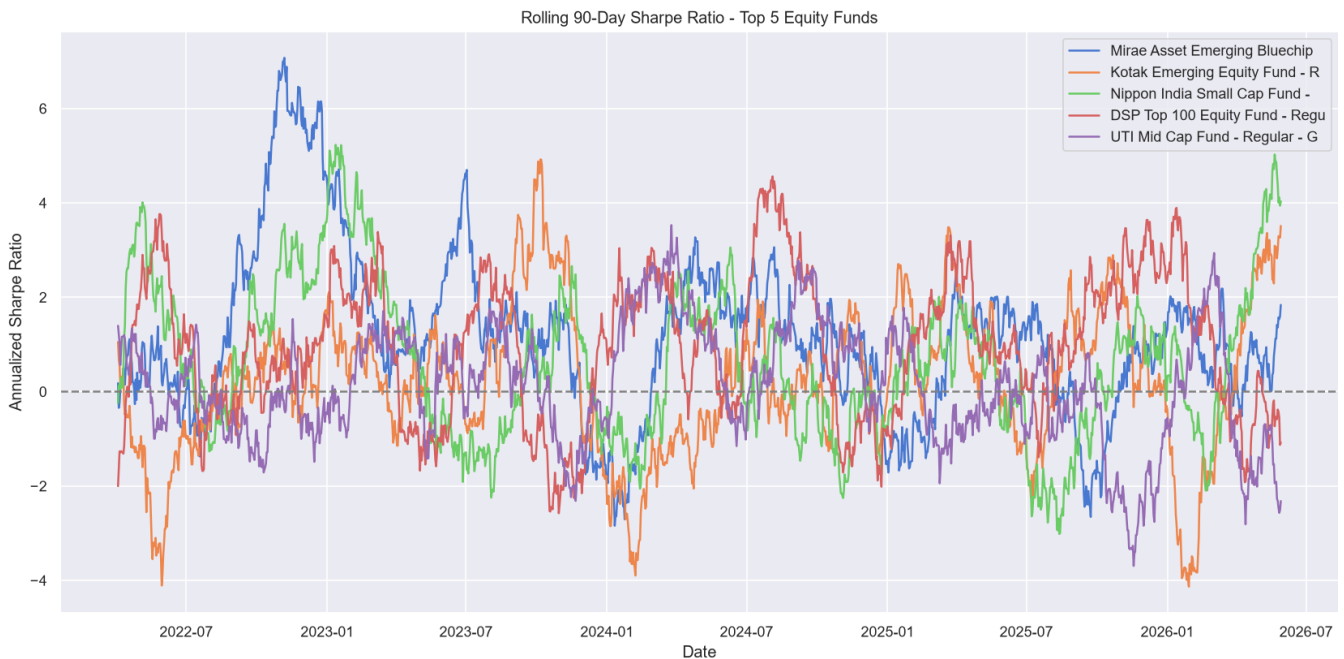


Tracking error calculations revealed that the top active funds successfully outperformed the passive indices over the 3-year horizon.

Advanced Analytics & Risk Metrics

Beyond standard metrics, we performed deep behavioral and risk modeling:

- Historical VaR (95%) and CVaR identified the funds with the most severe 'fat tail' risk during market corrections.
- 90-Day Rolling Sharpe highlighted that risk-adjusted performance is highly time-dependent.
- Investor Cohort Analysis mapped early vs late investor groups, proving earlier cohorts have significantly higher lifetime invested capital.
- SIP Continuity Analysis tracked investors with 6+ SIPs, flagging accounts with >35 days average gap as 'at-risk' of attrition.
- Sector HHI quantified portfolio concentration, showing high density in the Banking & Financial Services sectors.



Power BI Dashboard

The Interactive Power BI Dashboard comprises 4 comprehensive pages:

1. Overview: High-level KPI aggregations.
2. Fund Performance: Deep-dive into specific fund returns and scorecard ranks.
3. NAV Details: Historical time-series exploration.
4. SIP & Market Trends: Heatmaps and behavioral analytics.

[Note: Final dashboard deployment screenshots are maintained directly within the Power BI Service workspace.]

Key Findings & Recommendations

Key Findings:

1. The highest rated funds on our composite scorecard successfully generated positive Alpha against NIFTY 100.
2. Significant concentration risk (High HHI) exists in top funds.
3. Younger demographics (26-35) prefer SIPs, while higher-income bands execute large lumpsums.

Recommendations:

- Investors: Utilize the generated Scorecard to select funds based on risk-adjusted returns rather than just 1-year trailing absolute returns.
- Fund Managers: Monitor HHI metrics to ensure portfolios do not inadvertently become closet index trackers heavily overweight in Banking.
- Business/Product Teams: Target the 'at-risk' SIP investors (gap > 35 days) with automated retention campaigns.

Limitations & Conclusion

Limitations:

- Alpha and Beta were computed strictly against NIFTY 100 for all funds, which is a broad market comparison rather than category-specific.
- The 6.5% Risk-Free Rate is held constant historically, while in reality, repo rates fluctuated.
- Historical data limits the 5-year CAGR computation for newer funds.

Conclusion:

This capstone successfully demonstrates a production-grade analytics pipeline. By transforming raw CSV data into a unified relational database, and layering on Python-based quantitative modeling, we generated deep insights that bridge the gap between raw data and actionable financial intelligence.